Demonstration of Proposed Features

Date	3 July 2026
Team ID	XXXXXX
Project Name	**OptiCrop**: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features:

S.No	Feature Name	Description	Status	Demonstrated	Remarks
1	Crop Recommendation System	Predicts the most suitable crop based on soil nutrients and environmental conditions using a trained ML model.	Implemented	Yes	Core functionality successfully demonstrated.
2	Data Preprocessing Pipeline	Cleans the dataset, handles missing values/outliers, and performs feature scaling before training.	Implemented	Yes	Improved model accuracy and reliability.
3	Machine Learning Model Comparison	Trains and compares multiple algorithms (Logistic Regression, KNN, Decision Tree, Random Forest, K-Means) to select the best-performing model.	Implemented	Yes	Random Forest selected as the best model.
4	Web-Based User Interface	Interactive HTML/CSS/JavaScript interface for entering soil and climate parameters and displaying crop recommendations.	Implemented	Yes	Successfully integrated with the backend.
5	Flask Backend API	Processes user inputs, validates parameters, loads the trained model, and returns predictions.	Implemented	Yes	Prediction API tested successfully.
6	Real-Time Weather API Integration	Automatically fetch weather information to reduce manual input and improve recommendations.	Pending	No	Planned as a future enhancement.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	5
Total Features Demonstrated	5

Overall Implementation Rate (%)	83.3% (5/6 × 100)